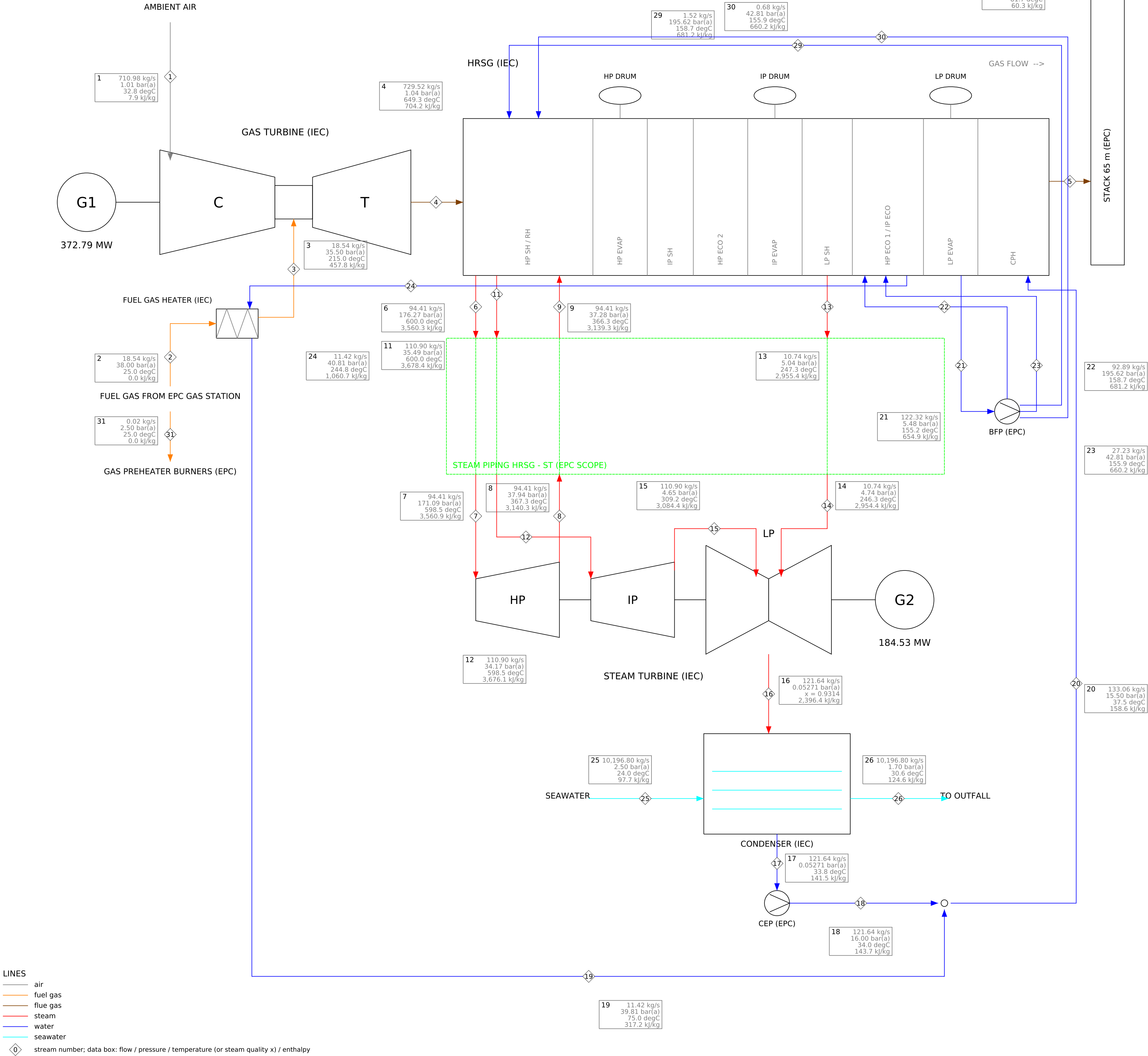




LINES
— air
— fuel gas
— flue gas
— steam
— water
— seawater
◇ stream number; data box: flow / pressure / temperature (or steam quality x) / enthalpy

STREAM TABLE - SUM-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	710.98	1.01	32.8	7.9	5.6
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	18.54	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	18.54	35.50	215.0	457.8	8.5
4	GT exhaust gas to HRSG inlet	729.52	1.04	649.3	704.2	513.7
5	Flue gas at HRSG outlet to stack	729.52	1.01	81.7	60.3	44.0
6	HP steam at HRSG outlet (HP SH outlet)	94.41	176.27	600.0	3,560.3	336.1
7	HP steam at ST main stop valve inlet	94.41	171.09	598.5	3,560.9	336.2
8	Cold reheat steam at HP turbine exhaust	94.41	37.94	367.3	3,140.3	296.5
9	Cold reheat steam at HRSG reheater inlet	94.41	37.28	366.3	3,139.3	296.4
10	IP superheated steam (joins cold reheat inside the HRSG)	15.81	37.99	328.3	3,043.5	48.1
11	Hot reheat steam at HRSG outlet	110.90	35.49	600.0	3,678.4	407.9
12	Hot reheat steam at ST IP stop/control valve inlet	110.90	34.17	598.5	3,676.1	407.7
13	LP steam at HRSG outlet	10.74	5.04	247.3	2,955.4	31.7
14	LP steam at ST LP admission	10.74	4.74	246.3	2,954.4	31.7
15	IP turbine exhaust (crossover, internal)	110.90	4.65	309.2	3,084.4	342.1
16	LP turbine exhaust to condenser (UEEP)	121.64	0.0527	x 0.931	2,396.4	291.5
17	Condensate at hotwell outlet / CEP suction	121.64	0.0527	33.8	141.5	17.2
18	Condensate at CEP discharge	121.64	16.00	34.0	143.7	17.5
19	Fuel gas heater water return to condensate	11.42	39.81	75.0	317.2	3.6
20	Condensate to HRSG condensate preheater inlet	133.06	15.50	37.5	158.6	21.1
21	Feedwater from LP drum to boiler feed pumps	122.32	5.48	155.2	654.9	80.1
22	HP feedwater at BFP discharge to HP economiser	92.89	195.62	158.7	681.2	63.3
23	IP feedwater at BFP interstage to IP economiser	27.23	42.81	155.9	660.2	18.0
24	IP economiser water to performance fuel gas heater	11.42	40.81	244.8	1,060.7	12.1
25	Circulating seawater to condenser inlet	10,196.80	2.50	24.0	97.7	995.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	30.6	124.6	1,270.2
29	HP attemper spray water (BFP discharge)	1.52	195.62	158.7	681.2	1.0
30	Reheat attemper spray water (BFP interstage)	0.68	42.81	155.9	660.2	0.4
31	Fuel gas to the station preheater burners (water-bath heaters)	0.02	2.50	25.0	0.0	0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	372.79	MW
ST output (generator terminals)	184.53	MW
Gross output	557.33	MW
Auxiliary loads	-9.47	MW
Transformer losses (GSU, UAT, MV/LV)	-1.65	MW
NET OUTPUT (HV terminals)	546.21	MW
Heat input (LHV)	881.16	MW
Gross heat rate (LHV)	5,686	kJ/kWh
NET HEAT RATE (LHV)	5,808	kJ/kWh
Net efficiency (LHV)	61.99	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	91,208	Sm ³ /h
Circulating water flow	36,169	m ³ /h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	88.97	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.002 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.002 MW (tolerance 1.29 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.001 MW (tolerance 0.89 MW)
OK	energy balance HRSG: residual +1.912 MW vs stated casing loss 1.879 MW (tolerance 0.91 MW)
OK	energy balance PIPE: residual +0.321 MW (tolerance 1.07 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.78 MW)
OK	CW temperature rise 6.61 K vs DP-CASE-0015 <= 7.0 K (Condenser cooling water temperature rise)
OK	gas flow 91,208 Sm ³ /h = 76.0 % of DP-OWNR-0033 (120,000 Sm ³ /h)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SUM-NG-100
CASE	Summer design (32.8 degC, 0.4 %), natural gas, base load
DOCUMENT	ALP-HMB-SUM-NG-100
STATUS	preliminary - generated by engine hmb v1.2.0
DATA	6b60fce553593bf5 2026-09-29
NOTE	all values calculated from the database; do not edit